COMP9900
Computer Science Project

Project: P30 - Policy in Action Library

9900-W11C-BREAD

Project Proposal

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1. Background

a. Problem Definition and Impact

1. Problem Definition
In an information-rich era marked by rapid data expansion and increasingly complex sources of information, quickly extracting key information from a complex domain remains a major challenge in solving complex problems. Public policy research and policy practice usually rely on a large volume of distributed documents, including policy research reports, government documents, academic papers, case studies, institutional publications, and private case materials accumulated within organisations. Although these materials are valuable and information-rich, they are often scattered across different platforms, formats, and access permissions. At the same time, the information is often highly homogeneous, and there is a lack of a unified, searchable, and continuously expandable knowledge repository. For policymakers, public sector professionals, researchers, and students, the real difficulty is not only "finding materials", but also quickly determining which materials are relevant to the current problem and extracting actionable experience, risks, implementation conditions, and case comparisons from multiple sources.

Existing research shows that decision-related evidence has long faced practical barriers in the policy process. Common problems encountered by policymakers when using research evidence include difficulty in obtaining high-quality and relevant research evidence in a timely manner (Oliver et al., 2014). At the same time, there remains a gap between the supply of research evidence and the needs of policy practice (OECD, 2020). Therefore, the core problem addressed by this project is not the complete absence of policy knowledge, but the difficulty of effectively organising, retrieving, understanding, and transforming such knowledge into practical support information.

In the context of the P30 Policy in Action Library project, this problem becomes more specific. The system needs to integrate two types of knowledge sources: publicly accessible policy research, policy reports, and case materials; and private case studies or internal teaching cases that may be uploaded later. This means that the platform cannot be merely a standard keyword search tool. Instead, it needs to support document upload, knowledge repository management, metadata tagging, semantic search, RAG synthesis, and source citation. Otherwise, even if users have access to a large amount of material, they will still struggle to obtain reliable answers efficiently.

2. Proposed Solution
Therefore, the core problem of this project can be defined as follows: how to build a source-grounded RAG knowledge repository system for policy case studies and policy research materials, enabling users to retrieve relevant documents through natural language questions and receive source-based synthesised responses. The

system does not attempt to automatically replace policymaking or public decision-making. Instead, it serves as a platform for supporting research and case-based learning, helping users understand existing materials more quickly, compare relevant cases, and extract practical lessons learned, risks, implementation considerations, and recommendations.

From a technical perspective, Retrieval-Augmented Generation (RAG) is suitable for this task because it combines the generative capability of large language models with retrieval from an external knowledge repository. RAG methods can retrieve external documents to handle knowledge-intensive NLP tasks and improve the specificity and factuality of generated content (Lewis et al., 2020). Later surveys of RAG also indicate that RAG helps integrate domain knowledge, support knowledge updates, and improve the credibility of generated results (Gao et al., 2023). This is particularly important for this project because policy materials will continue to expand, and user private cases may also be gradually added at later stages of the project.

3. Impact

The expected impact of the Policy in Action Library is to reduce the cost for users to obtain key insights from distributed policy materials, improve the efficiency of policy case learning and research retrieval, and create a unified knowledge management entry point for public resources and private client cases. Through natural language queries, semantic search, and synthesised responses with source references, the system can help users find relevant cases more quickly, understand lessons learned from similar policy challenges, and identify potential risks and implementation considerations.

At the current prototype stage, this impact is mainly reflected in improvements to information access efficiency and research workflows. For policymakers or public sector practitioners, the system can serve as an initial decision-support research tool, helping them form an overall understanding of relevant materials before policy design, implementation evaluation, stakeholder engagement, or case-based learning. For researchers and students, the system can reduce the time spent on repetitive searching and manual organisation of materials, allowing them to focus more on case comparison, evidence interpretation, and analysis of policy implications.

From a future development perspective, the Policy in Action Library can gradually expand into a continuously accumulated knowledge repository for policy practice. As public materials and private case studies continue to be added, the system can not only answer individual questions, but also help users identify recurring themes, common implementation risks, and transferable lessons across cases. If more fine-grained metadata, access control, and evaluation feedback mechanisms are added in later stages, the platform has the potential to support longer-term institutional knowledge management and help policymakers obtain more transparent and traceable reference evidence when dealing with complex policy problems. Since AI systems in public sector contexts should pay particular attention to transparency, accountability, and knowledge base control, this source-traceable design can also help reduce the risk of users blindly trusting generative AI outputs (Carullo, 2023).

Therefore, the value of this project does not lie in "making policy" on behalf of users. Rather, it lies in helping users obtain traceable, explainable, and actionable support information from existing policy knowledge more reliably and efficiently. In other words, this project should be positioned as a NotebookLM-like RAG knowledge platform for policy research and case-based learning, rather than an automated policymaking or decision-making system.

References

Carullo, G. (2023). Large language models for transparent and intelligible AI-assisted public decision-making. CERIDAP, 2023(3), 1-23. https://doi.org/10.13130/2723-9195/2023-3-100
Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey. arXiv. https://arxiv.org/abs/2312.10997
Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W., Rocktaschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. Advances in Neural Information Processing Systems, 33, 9459-9474. https://arxiv.org/abs/2005.11401
OECD. (2020). Building capacity for evidence-informed policy-making: Lessons from country experiences. OECD Publishing. https://doi.org/10.1787/86331250-en
Oliver, K., Innvar, S., Lorenc, T., Woodman, J., & Thomas, J. (2014). A systematic review of barriers to and facilitators of the use of evidence by policymakers. BMC Health Services Research, 14, Article 2. https://doi.org/10.1186/1472-6963-14-2

b. Review of Existing Systems

The core of this project is to build a source-grounded RAG knowledge base for policy case studies and policy research materials by drawing on relevant technologies from existing AI knowledge base platforms and policy data. Therefore, the review of existing systems can be conducted from three complementary perspectives: first, AI knowledge base platforms that employ similar methods but utilise different data; second, policy databases with similar subject matter but non-natural language interaction methods;third, enterprise-level RAG platforms with similar engineering architectures but more general domain objectives.

1. NotebookLM / IMA-type AI knowledge base platforms

1.1 Problems Addressed by the Systems

NotebookLM and IMA are two AI knowledge base platforms similar to the methodology of this project. The primary problem addressed by such platforms is that of users who possess a large volume of material but find it difficult to read, organise and formulate questions about it quickly.NotebookLM allows users to interact with AI based on uploaded or added sources, and supports a variety of source types including Google Docs, PDFs, text, web URLs, YouTube videos and audio; Google's documentation also states that NotebookLM's responses utilise sources selected or provided by the user as context (Google, n.d.-a; Google, n.d.-b). IMA, meanwhile, provides processes for knowledge base creation, file import, information processing and indexing, and shared knowledge base queries, illustrating that such platforms have established a workflow for 'data import – knowledge base management – AI Q&A' tailored for general users (Hong Kong Metropolitan University, n.d.).

1.2 Design Implications

From a user perspective, the most valuable advantage of such systems is 'source-grounded question answering'. They first allow users to provide or select sources, and then generate answers based on this material. Considering the specific details of this project, policy users need to know which case studies, reports or research materials the answers are derived from, rather than merely seeing conclusions generated by the model; this approach therefore meets their requirements;Secondly, the low barrier to entry for natural language question-answering on AI-style knowledge base platforms—such as NotebookLM/IMA—means that users do not need to understand embeddings, chunking or vector databases; they simply need to ask questions in natural language. This inspires us to abstract the complex backend retrieval process in an object-oriented manner, enabling users to search and query relevant literature in the database using natural language, thereby allowing policymakers or users to utilise the system in a question-driven manner.

Consequently, this project can draw upon the proven architecture of NotebookLM/IMA—'build the knowledge base first, then ask questions about the sources'—and proceed to adapt it for our specific domain.Specifically, this involves retaining document upload, source selection, natural language querying and source citation, whilst adding policy-specific metadata, a distinction between public and private sources, admin upload and management, and response templates tailored to policy practice. In other words, the core technical design can draw upon a general-purpose NotebookLM system platform to build a policy-focused NotebookLM-like RAG prototype based on a relatively mature architecture.

1.3 Limitations to be Overcome

The main limitations of NotebookLM and IMA lie in the fact that they are general-purpose knowledge platforms, rather than domain-specific systems designed for policy case learning. They typically do not embed policy metadata—such as policy area, jurisdiction, sector, stakeholder type and implementation challenges—within their knowledge bases, nor do they organise responses by default into policy examples, lessons learnt, risks and implementation considerations.Furthermore, NotebookLM imposes limits on the number of sources, the number of words per source, and file size; Google's FAQ also states that if source content is too brief, the system may simply cite the entire document rather than providing a more specific citation (Google, n.d.-b). These characteristics indicate that general-purpose AI knowledge platforms are suitable for inspiring interaction methods, but cannot directly meet this project's requirements for the organisation of policy materials and practice-oriented, comprehensive responses.

2. Azure AI Search

2.1 Problems Addressed by the System

Azure AI Search represents an enterprise-grade RAG platform with an architectural structure similar to that of this project. It addresses the following problem: organisations typically possess large volumes of private, dynamic, unstructured or semi-structured data, which general-purpose LLMs cannot reliably access directly.Azure AI Search describes RAG as a pattern in which a retrieval system grounds LLM responses in user content, and provides capabilities such as classic RAG, agentic retrieval, semantic ranking, OCR, PDF/image extraction and access control (Microsoft, n.d.).

2.2 Design Insights

The most significant advantage of such platforms is that they demonstrate the engineering architecture of a mature RAG system. They break RAG down into several distinct steps: data ingestion, document parsing, chunking, embedding, indexing, retrieval, reranking, answer generation and source attribution. System design based on this framework is highly valuable, as although our project is a prototype, it must ensure that data flows and the basis for answers are clear and explainable, whilst successfully utilising natural language to retrieve relevant resources.

This naturally leads to the following insights:

1. RAG pipeline structure: Azure AI Search emphasises chunking, vectorisation, semantic ranking and hybrid retrieval. This inspires us to design the backend as distinct ingestion and query pipelines.
2. Source attribution: Amazon Bedrock explicitly supports the inclusion of citations in generated responses, enabling users to verify the original data sources. Similarly, the citations provided in our project should not merely consist of a references page, but should display the relevant source within each response.
3. Access control and private data: The Azure AI Search documentation emphasises access control at both the knowledge source and document levels. This means that the distinction between public materials and private client cases should be incorporated into the design assumptions, and access and upload permissions should be configured to enforce access control.

2.3 Limitations to be Overcome

The limitation of platforms such as Azure AI Search is that they are general-purpose, enterprise-grade platforms: whilst they offer robust engineering capabilities, they are characterised by high complexity and cost. They do not provide a metadata schema tailored to policy case learning, nor do they determine whether the responses received by users should include lessons learnt, risks or implementation considerations.For this course project, a direct replication of an enterprise-grade RAG platform could introduce excessive cloud service configuration, permission systems, OCR pipelines, cost control issues and vendor lock-in, exceeding the reasonable scope of a prototype. For this course project, a lightweight engineering approach should be adopted for such platforms, rather than a full-scale replication of enterprise-grade products. The MVP should prioritise support for the upload of text-based PDFs or text documents, metadata tagging, chunking, semantic retrieval, RAG answer generation and source citation.Complex chart interpretation, fine-grained access control, automated web crawling and advanced re-ranking can be treated as future work or stretch features. This approach ensures engineering soundness whilst avoiding the expansion of the project scope beyond the specifications of a production-grade system.

3. Review of Existing Systems

In summary, both of the existing systems described above can serve as design references for this project, but neither fully meets its combined requirements. AI knowledge base platforms such as NotebookLM/IMA demonstrate the interactive capabilities of source-grounded AI knowledge bases, but lack policy-domain metadata and practice-oriented outputs. Azure AI Search showcases the engineering pipeline, access control and source attribution of a mature RAG system, but as a general-purpose enterprise platform, it is overly complex and does not provide a policy-specific knowledge structure.Consequently, this project should be positioned as a lightweight, policy-domain-specific RAG prototype: drawing on curated public policy documents and private client case studies as knowledge sources, it supports administrative management and metadata tagging, generates comprehensive, cited responses via natural language queries, and focuses its outputs on policy examples, lessons learnt, risks, implementation considerations and recommendations, thereby reducing.

References

Google. (n.d.-a). NotebookLM. https://notebooklm.google/

Google. (n.d.-b). Frequently asked questions – NotebookLM Help. https://support.google.com/notebooklm/answer/16269187

Hong Kong Metropolitan University. (n.d.). ima.copilot. https://www.hkmu.edu.hk/oetools/imacopilot/

Microsoft. (n.d.). RAG and generative AI in Azure AI Search. https://learn.microsoft.com/en-us/azure/search/retrieval-augmented-generation-overview

2: User Stories and Sprints

a. Scope Statement

The project will ultimately deliver a prototype AI database for public policy. This is an artificial intelligence platform based on a professional open-source knowledge base for public policy, helping students and public policy researchers understand relevant research and specific cases, and providing professional analysis and reference suggestions. The project will focus on selecting and building the public policy knowledge base and AI question-answering process. Production deployment, vertical LLM training, and model interpretability are not within the scope of this prototype.

JZ Jim Zha

答复 全部答复 转发

收件人: Jessica Genauer; Rebecca Razavi; Xinyu Chen; Yinjie Ren; + 另外 3 人

周五 2026/6/12 12:1

此消息的语言为 英语

翻译至 中文 (简体) 始终不翻译 英i

Dear Jessica and Reebecca,

Thank you again for meeting with us today. It was very helpful for us to better understand the direction and expectations for the Policy in Action Library project.

I noticed that I do not currently have permission to download the Teams transcript or written record of the meeting. If possible, could you please share the transcript with us or enable access to it? This would help us keep accurate meeting minutes and ensure our project proposal reflects your expectations correctly.

In the meantime, we have summarised our understanding of the key points from today's discussion below:

1. Project direction
   The project should focus on building a prototype of a Policy in Action Library that helps users access, search, and synthesise policy case studies, policy reports, and eventually peer-reviewed academic research.
2. Reference platforms
   Useful existing examples include:

- Harvard Kennedy School Case Program
- Columbia's Case Collection

JG Jessica Genauer

答复

收件人: Jim Zha; Rebecca Razavi; Xinyu Chen; Yinjie Ren; Zhicheng Hu; Changyu Chen; Zhanpeng Zhu

此消息的语言为 英语

Thanks Jim,

That's a great summary.

Does this link work?

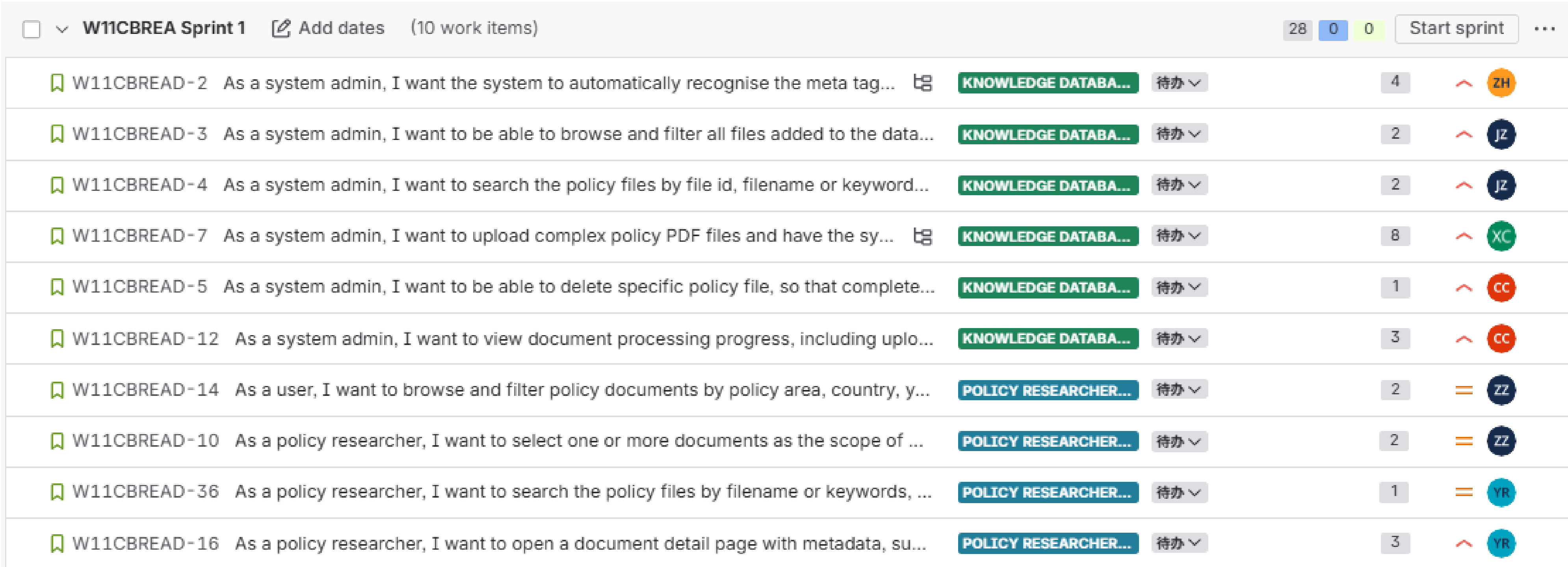
Recap: COMP9900 project 12 June | Meeting | Microsoft Teams

Best,
Jessica

***

Dr. Jessica Genauer
Academic Director of the Public Policy Institute
Associate Professor in International Relations
University of New South Wales (UNSW)

b. Sprint 1 User Stories



W11CBREA Sprint 1 Add dates (10 work items) 28 0 0 Start sprint

W11CBREAD-2	As a system admin, I want the system to automatically recognise the meta tag...	KNOWLEDGE DATABA...	待办	4	^	ZH
W11CBREAD-3	As a system admin, I want to be able to browse and filter all files added to the data...	KNOWLEDGE DATABA...	待办	2	^	JZ
W11CBREAD-4	As a system admin, I want to search the policy files by file id, filename or keyword...	KNOWLEDGE DATABA...	待办	2	^	JZ
W11CBREAD-7	As a system admin, I want to upload complex policy PDF files and have the sy...	KNOWLEDGE DATABA...	待办	8	^	XC
W11CBREAD-5	As a system admin, I want to be able to delete specific policy file, so that complete...	KNOWLEDGE DATABA...	待办	1	^	CC
W11CBREAD-12	As a system admin, I want to view document processing progress, including uplo...	KNOWLEDGE DATABA...	待办	3	^	CC
W11CBREAD-14	As a user, I want to browse and filter policy documents by policy area, country, y...	POLICY RESEARCHER...	待办	2	=	ZZ
W11CBREAD-10	As a policy researcher, I want to select one or more documents as the scope of ...	POLICY RESEARCHER...	待办	2	=	ZZ
W11CBREAD-36	As a policy researcher, I want to search the policy files by filename or keywords, ...	POLICY RESEARCHER...	待办	1	=	YR
W11CBREAD-16	As a policy researcher, I want to open a document detail page with metadata, su...	POLICY RESEARCHER...	待办	3	^	YR

Total story points: 86
Sprint1 story points: 28

Num	Priority	User Story	User Acceptance Criteria
2	High	As a system admin, I want the system to automatically recognise the meta tags for a file after it has been uploaded, and allow me to maunally select or edit these tags, so that later on policy researchers can filter files precisely based on these tags.	After the system completes file processing, it can automatically add relevant meta tags to the file, such as the country/region of origin and year of the policy. The admin can manually tag uploaded files.
3	High	As a system admin, I want to be able to browse and filter all files added to the database, so that I can have an overview of files and ensure files have been successfully processed into their respective subcategories.	The admin can clearly see all uploaded files. The admin can filter files by various tags.
4	High	As a system admin, I want to search the policy files by file id, filename or keywords, so that I can quickly and accurately locate specific document(s).	The admin can search for corresponding files using information such as ID and name. If no relevant file is found, return files with similar tags.
7	High	As a system admin, I want to upload complex policy PDF files and have the system automatically perform semantic segmentation, so that when a policy researcher asks the AI, it can retrieve the full unbroken context and provide an accurate response.	The system should have the function of properly chunking uploaded PDF files. Each chunk should maintain contextual relationships through tags such as documentID, section, and page.
5	High	As a system admin, I want to be able to delete specific policy file, so that completely erasing the document from search index and preventing policy researchers and the AI from continuing to reference incorrect or outdated contents.	The admin can delete files from the database. The system should require secondary confirmation during deletion. After deletion, the AI search results will not include the file.
12	High	As a system admin, I want to view document processing progress, including upload, parsing, segmentation, embedding, and indexing status, so that I know whether a document is ready for retrieval.	Each file should display its processing status, and if it is being processed, a progress bar or other visual system should be provided. If the process fails, an error message should be returned.
14	Mid	As a user, I want to browse and filter policy documents by policy area, country, year, source, and tags, so that I can discover relevant cases before asking questions.	Users can select various document types using tags to provide a basis for AI responses.

10	Mid	As a policy researcher, I want to select one or more documents as the scope of my question, so that the AI response is focused on relevant sources.	Users can select one or more documents. After a user submits a question, the system only retrieves the selected documents.
36	Mid	As a policy researcher, I want to search the policy files by filename or keywords, so that I can quickly and accurately locate specific document(s).	The User can search for corresponding files using information such as ID and name. If no relevant file is found, return files with similar tags.
16	High	As a policy researcher, I want to open a document detail page with metadata, summary, source information, and available snippets, so that I can decide whether the document is relevant before asking the AI.	Users can access the details page from the document list or search results. The details page should display the title, tags, source, year, summary, and available excerpts.

c. Product Backlog

Backlog (18 work items) 58 0 0 Create sprint

Key	Summary	Epic	Status	Story points
W11CBREAD-22	As an admin, I want to register trusted official websites, so that the system can collect policy documents from approved sources.	KNOWLEDGE DATABA...	待办	2
W11CBREAD-23	As an admin, I want to monitor website crawling status, so that I know whether new documents have been successfully added to the knowledge base.	KNOWLEDGE DATABA...	待办	4
W11CBREAD-1	As a system admin, I want to be able to upload multiple policy PDF files, so that I can efficiently expand the policy database	KNOWLEDGE DATABA...	待办	1
W11CBREAD-8	As a system admin, I want to log in securely by entering my username and password, and the system to block all unauthorised users, so that the system database is sec...	KNOWLEDGE DATABA...	待办	2
W11CBREAD-9	As a policy researcher, I want to ask natural-language questions about selected policy documents, so that I can quickly obtain evidence-based policy insights.	POLICY RESEARCHER...	待办	8
W11CBREAD-15	As a policy researcher, I want to compare policy cases across different countries or regions, so that I can understand differences in policy design and outcomes.	POLICY RESEARCHER...	待办	5
W11CBREAD-11	As a policy researcher, I want the AI response to include citations and source snippets, so that I can verify the evidence behind the answer.	POLICY RESEARCHER...	待办	4
W11CBREAD-13	As a user, I want the system to tell me when there is insufficient evidence in the current document library, so that I do not mistake unsupported AI output for fact.	POLICY RESEARCHER...	待办	4
W11CBREAD-19	As a policy researcher, I want to click citations in an AI response and view the supporting source snippet, so that I can verify the evidence without leaving the answer p...	POLICY RESEARCHER...	待办	3
W11CBREAD-20	As a policy researcher, I want to rate whether an AI answer is helpful and whether its citations are useful, so that the system can improve retrieval and answer quality o...	POLICY RESEARCHER...	待办	2
W11CBREAD-18	As a policy researcher, I want to see clear loading, retrieval, and answer generation states, so that I understand what the system is doing while waiting for an AI respon...	POLICY RESEARCHER...	待办	2
W11CBREAD-24	As a policy researcher, I want the AI response to be structured into relevant cases, key lessons, risks, implementation considerations, and practical recommendations, s...		待办	4
W11CBREAD-26	As a policy student, I want the AI to explain policy cases in clear, beginner-friendly language, so that I can understand complex policy concepts without needing extens...		待办	4
W11CBREAD-34	As a policy student, I want the system to generate a short learning summary of a selected policy document, including key concepts, main arguments, and important poli...		待办	4
W11CBREAD-35	As a policy student, I want the system to provide a comparison between two selected policy cases in terms of context, policy approach, outcomes, and lessons learned...		待办	4
W11CBREAD-27	As a policy student, I want the system to provide guided follow-up questions after an AI answer, so that I can continue learning about related policy issues, implementa...		待办	3
W11CBREAD-25	As a policy researcher, I want to export or save an AI-generated answer together with its citations and selected source documents, so that I can reuse the result in rep...		待办	2
W11CBREAD-17	DELETED-Duplicate with us10. As a policy researcher, I want to see and manage my selected documents in a visible document basket, so that I can clearly control whic...	POLICY RESEARCHER...	DELETED	0

Num	Priority	User Story	User Acceptance Criteria
22	Low	As an admin, I want to register trusted official websites, so that the system can collect policy documents from approved sources.	Administrators can add, edit, and delete trusted websites. The system can only collect or retrieve policy information from websites approved by the admin.
23	Low	As an admin, I want to monitor website crawling status, so that I know whether new documents have been successfully added to the knowledge base.	Administrators can see the crawling status of each website. After successful crawling, the new document should appear in the knowledge base, awaiting processing.
1	Lowest	As a system admin, I want to be able to upload multiple policy PDF files, so that I can efficiently expand the policy database.	Admin can select valid PDF files to upload into the database. The uploaded file should include information such as file size and upload time.
8	Low	As a system admin, I want to log in securely by entering my username and password, and the system to block all unauthorised users, so that the system database is secured from tampering by malicious users and spam.	It has a normal login system Unregistered users cannot access sensitive pages, such as database management pages.
9	Mid	As a policy researcher, I want to ask natural-language questions about selected policy documents, so that I can quickly obtain evidence-based policy insights.	Users can ask questions using natural language. The system's response should be based on specific files in the database.
15	Mid	As a policy researcher, I want to compare policy cases across different countries or regions, so that I can understand differences in policy design and outcomes.	Users can choose two or more policy cases for comparison. The comparison should include the corresponding source.
11	Low	As a policy researcher, I want the AI response to include citations and source snippets, so that I can verify the evidence behind the answer.	Information in AI responses needs to be accompanied by citations. The citation displays the document title, relevant excerpts, and source information. It also allows users to jump to the document.
13	Mid	As a user, I want the system to tell me when there is insufficient evidence in the	When no relevant content is found, the system should indicate that the database

		current document library, so that I do not mistake unsupported AI output for fact.	information is insufficient. The system should not force a definitive answer when there is insufficient evidence.
19	Mid	As a policy researcher, I want to click citations in an AI response and view the supporting source snippet, so that I can verify the evidence without leaving the answer page.	After a user clicks on a citation, the system should display the corresponding source snippet. The snippet should include the document title and related text.
20	Lowest	As a policy researcher, I want to rate whether an AI answer is helpful and whether its citations are useful, so that the system can improve retrieval and answer quality over time.	Users can rate the quality of the answers. Users can rate the citation usefulness. These ratings are returned to the admin as data to help resolve issues and iterate the system.
18	Mid	As a policy researcher, I want to see clear loading, retrieval, and answer generation states, so that I understand what the system is doing while waiting for an AI response.	The system should provide clear visual information. After a user submits a question, it should display "retrieving sources." Once the search is complete, it should display "generating answer."
24	Mid	As a policy researcher, I want the AI response to be structured into relevant cases, key lessons, risks, implementation considerations, and practical recommendations, so that I can quickly turn policy documents into actionable insights.	AI responses should include a fixed structure or clear headings. Each section should be generated based on the retrieved sources as much as possible.
26	Mid	As a policy student, I want the AI to explain policy cases in clear, beginner-friendly language, so that I can understand complex policy concepts without needing extensive prior knowledge.	Users can choose from several different response styles, such as beginner mode/student mode. In this mode, the AI's responses will be simple, clear, and easy to understand, maintaining a lively language style as long as there are no factual errors.
34	Mid	As a policy student, I want the system to generate a short learning summary of a selected policy document, including key concepts, main arguments, and important policy terms, so that I can understand the document before reading it in full.	After selecting a document, students can generate a learning summary. The summary should include key concepts, main arguments, and important terms. The summary should be based on the content of the selected document and source.

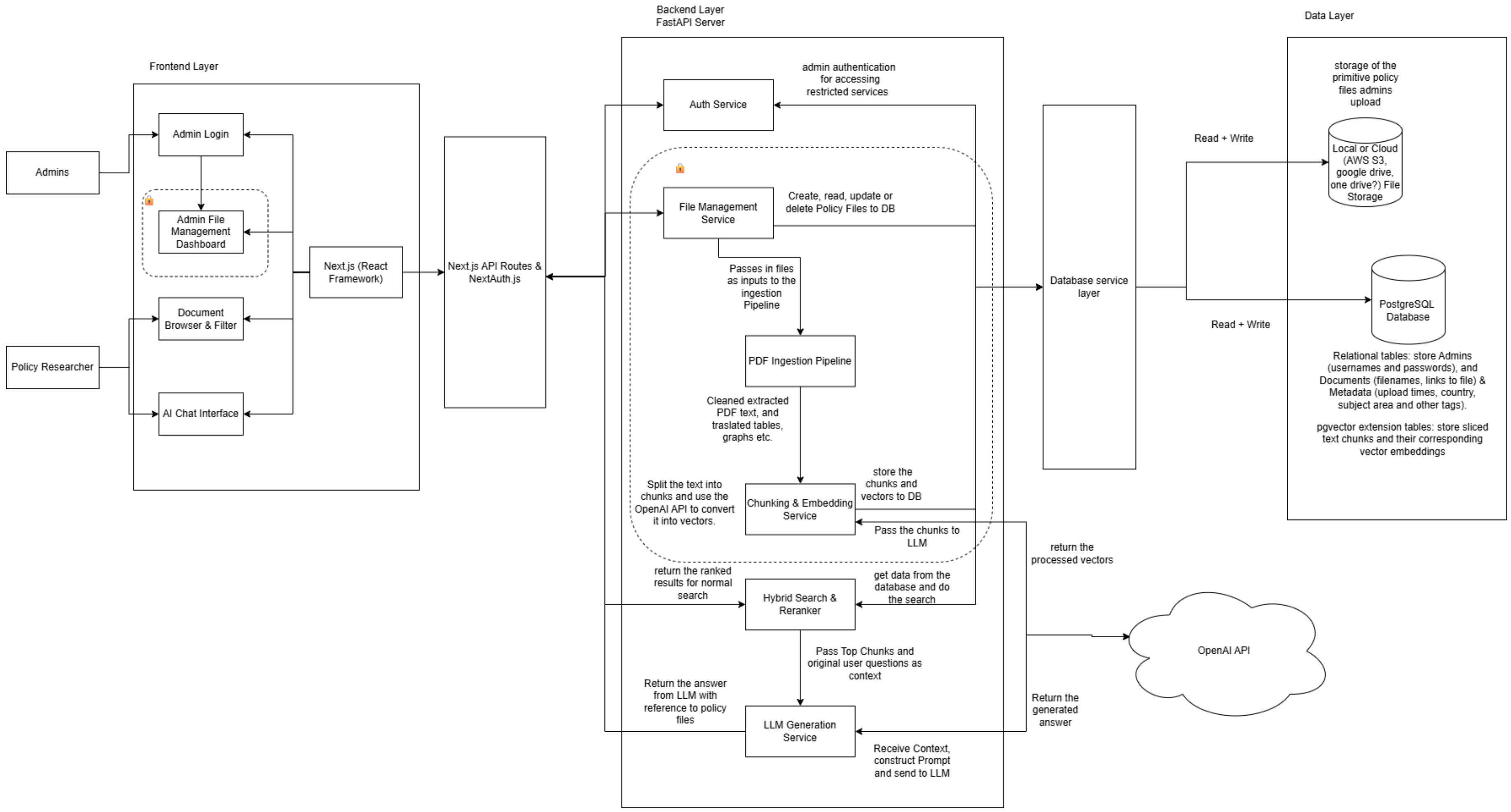
35	Mid	As a policy student, I want the system to provide a comparison between two selected policy cases in terms of context, policy approach, outcomes, and lessons learned, so that I can learn how different policy decisions lead to different results.	Students may choose two policy cases for comparison. The comparison should be presented in language appropriate for student comprehension and include source citations.
27	Mid	As a policy student, I want the system to provide guided follow-up questions after an AI answer, so that I can continue learning about related policy issues, implementation challenges, and real-world examples.	The answer can be followed by guiding questions to help students advance their learning progress.
25	Mid	As a policy researcher, I want to export or save an AI-generated answer together with its citations and selected source documents, so that I can reuse the result in reports, presentations, or further policy research.	The system can export AI question-and-answer records, as well as related documents, including citations and other necessary information.

d. Sprint Timeline and Milestones

Our project will be divided into three sprints according to the course requirements. Sprint 1 will be responsible for the design and construction of the core knowledge base system, sprint 2 for the design and construction of the user interaction system, and sprint 3 for refining details, project testing, documentation, and final demonstration.

Sprint	Timeline	Main Focus	Key Milestones and Deliverables
1	Week 3 Lab Time→ Week 5 Lab Time	Core knowledge database and document processing foundation	Define the project's service scope, establish the database architecture, implement PDF upload functionality, implement meta-tag tagging functionality, implement document browsing and retrieval functionality, and implement a basic RAG system, such as document parsing, chunking, and relevant contextual linking tags. Run the entire AI question-answering process successfully. Prepare a basic demo demonstrating how the system stores and processes uploaded files and databases them.
2	Week 5 Lab Time → Week 8 Lab Time	Main AI retrieval and researcher/student-facing workflow	A user-friendly UI was built, allowing natural language input via a chat window. Users can select document ranges, search for relevant passages, and generate AI responses including sources and citations. Users can directly access the database library to view the content they need, and can find relevant policy content through tags. Prepare a progressive demo showing the complete query-to-answer workflow.
3	Week 8 Lab Time → Week 10 Lab Time	Polish, testing, evaluation, and final delivery	Improve UI clarity and loading states. Add insufficient-evidence handling; refine citation display. Test main user workflows; fix bugs; complete technical documentation, user guide, and final report. Prepare the final demonstration and code submission.

3. Technical Design (a. System Architecture Diagram)

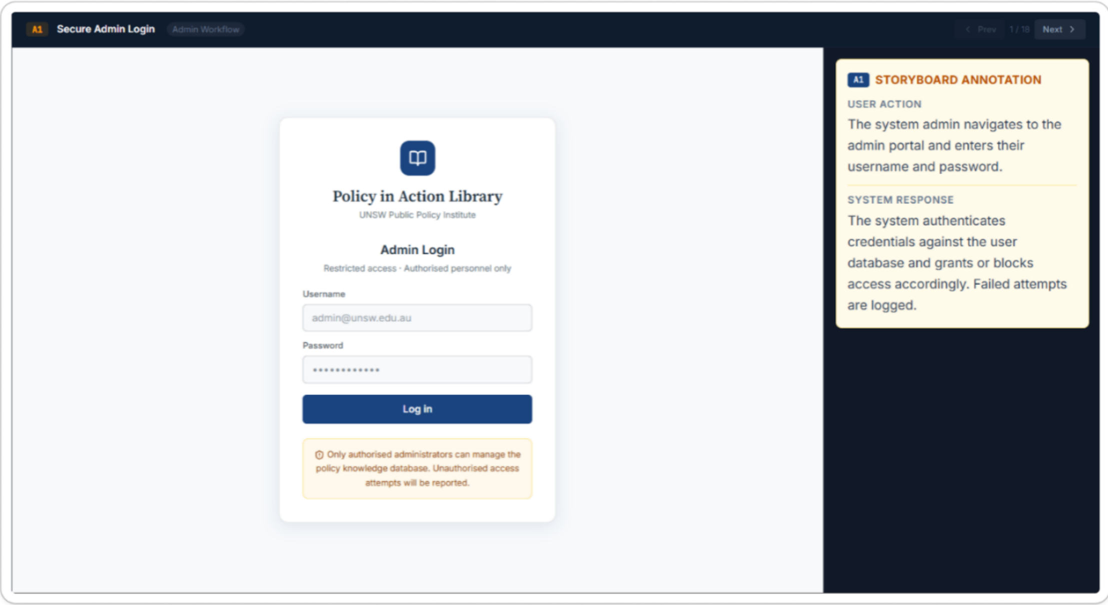


b. Interface storyboarding

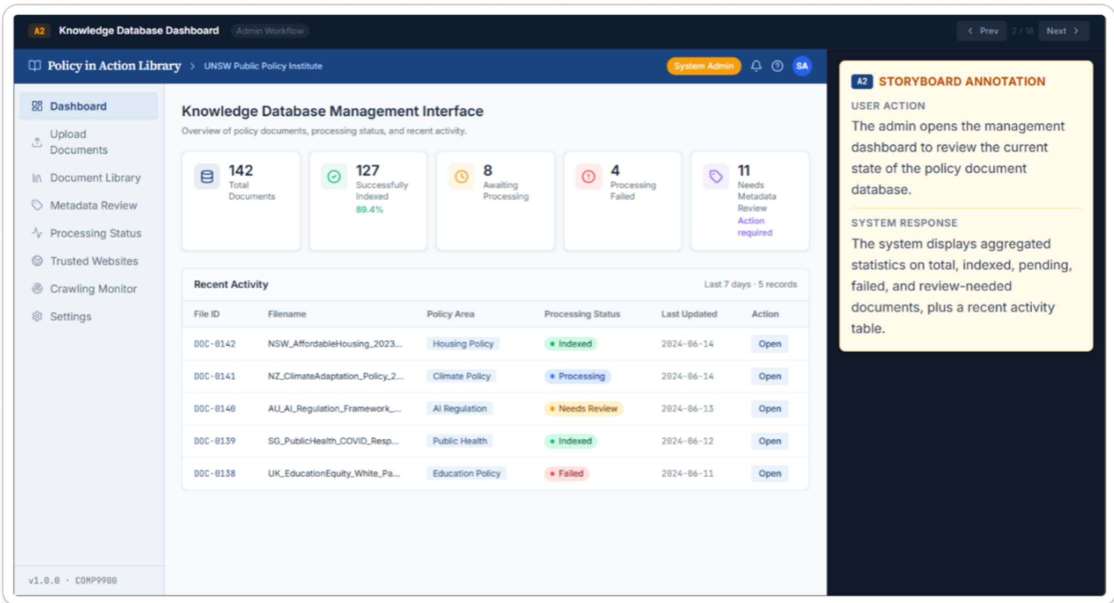
1 Administrator Workflow Storyboards

1.1 Administrator Login and Dashboard Access

Covered User Story: W11CBREAD-8



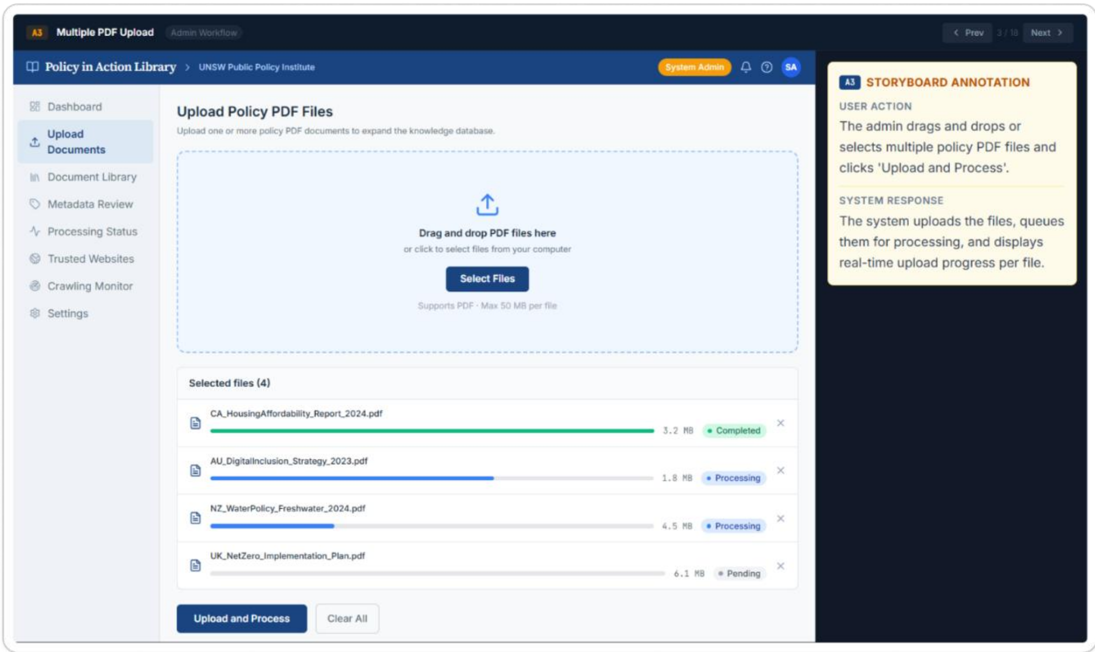
Log in successfully



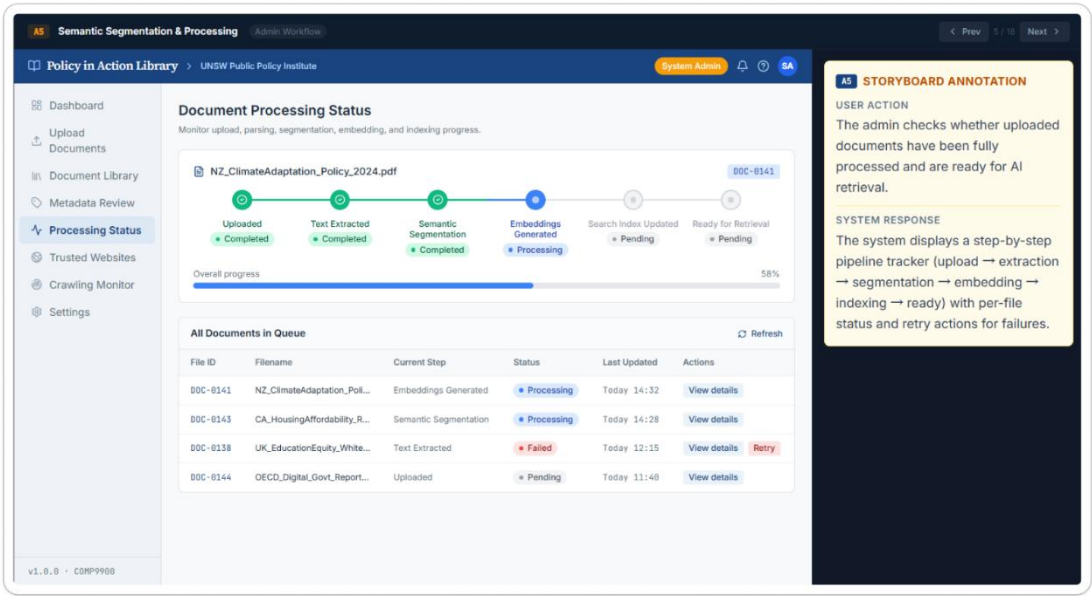
Demonstration: After entering valid login details, the administrator is taken to the dashboard. From there, they can view recent document activity and access the main administration functions (**W11CBREAD-8**).

1.2 PDF Upload and Processing

Covered User Stories: W11CBREAD-1, W11CBREAD-7, W11CBREAD-12



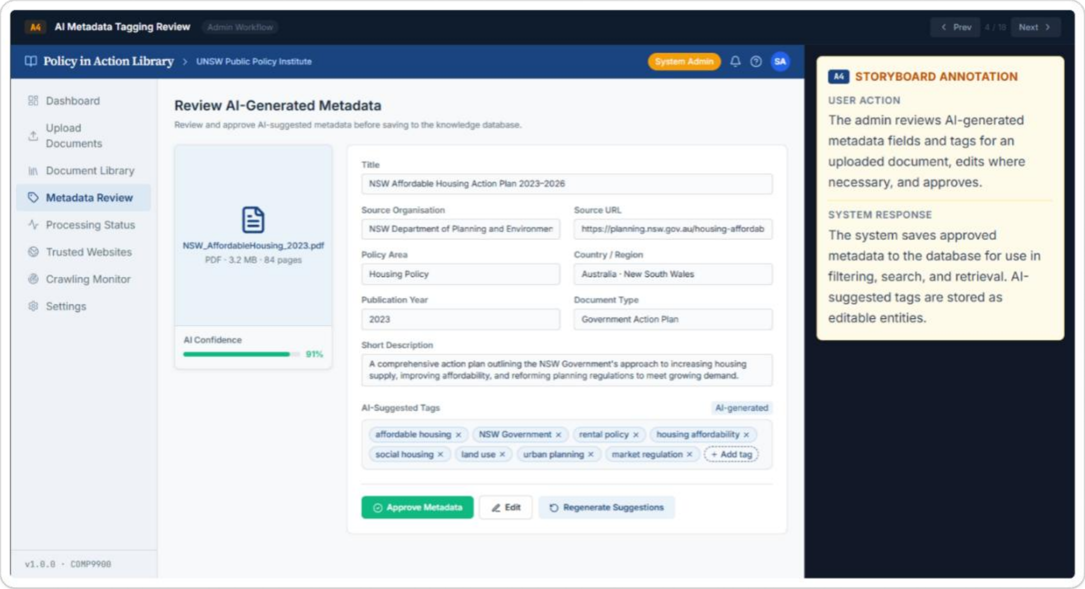
Select PDF files and click "Upload and Process"



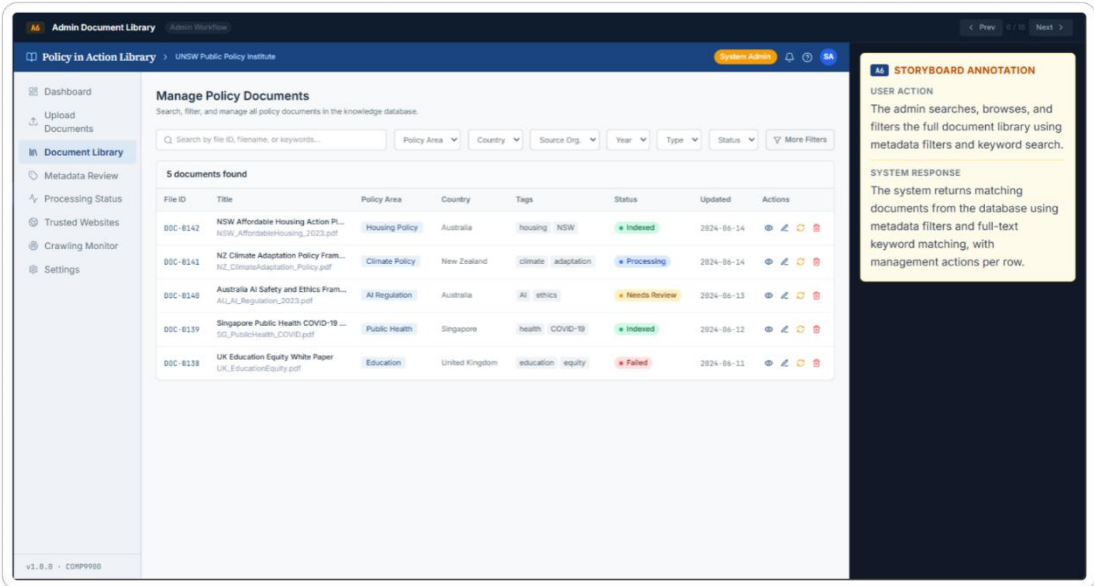
Demonstration: Administrators can upload several policy PDFs at the same time (**W11CBREAD-1**). The system extracts and divides the document content into meaningful segments for later retrieval (**W11CBREAD-7**). Processing progress is shown for each file, including extraction, segmentation, embedding and indexing. Failed tasks can also be retried (**W11CBREAD-12**).

1.3 Metadata Review and Approval

Covered User Story: W11CBREAD-2



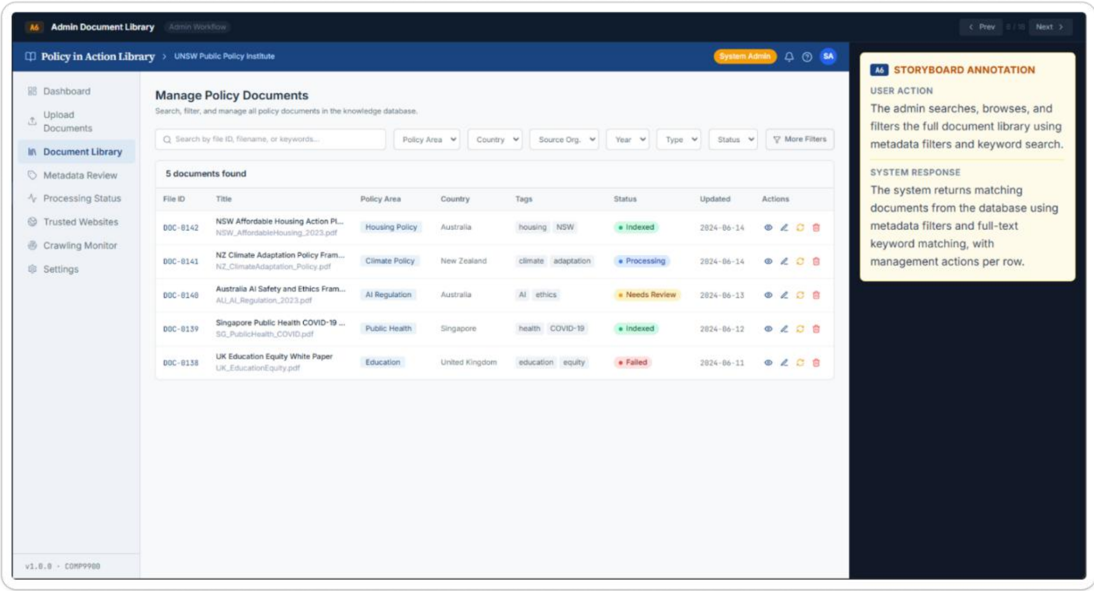
Review the generated metadata and click “Approve Metadata”



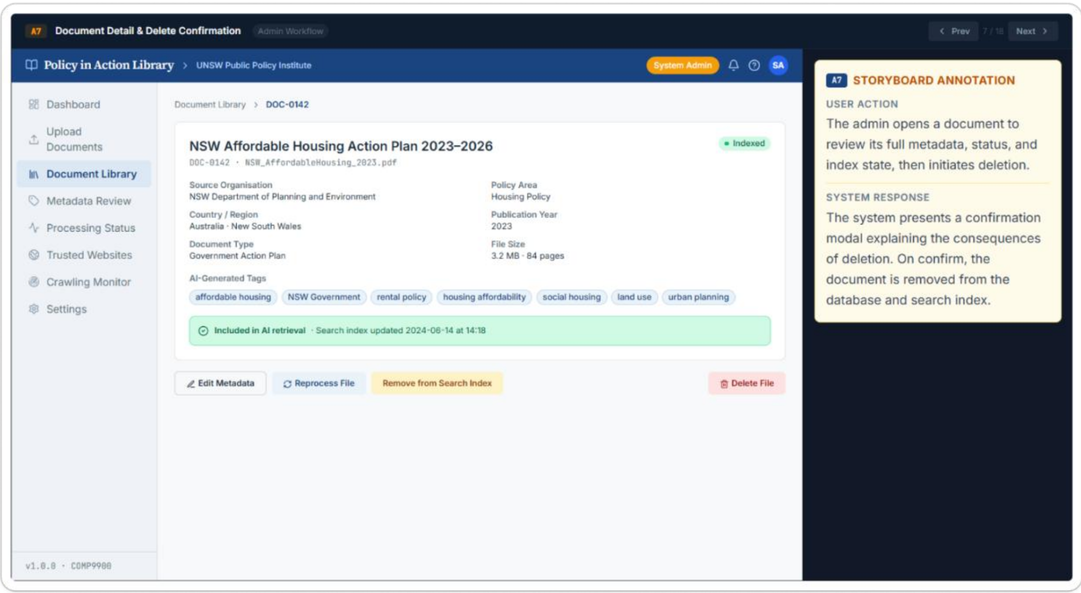
Demonstration: Once a document has been processed, the system suggests relevant metadata. The administrator can review and edit these fields before approval. Only approved documents are added to the searchable policy library (**W11CBREAD-2**).

1.4 Document Search and Management

Covered User Stories: W11CBREAD-3, W11CBREAD-4, W11CBREAD-5



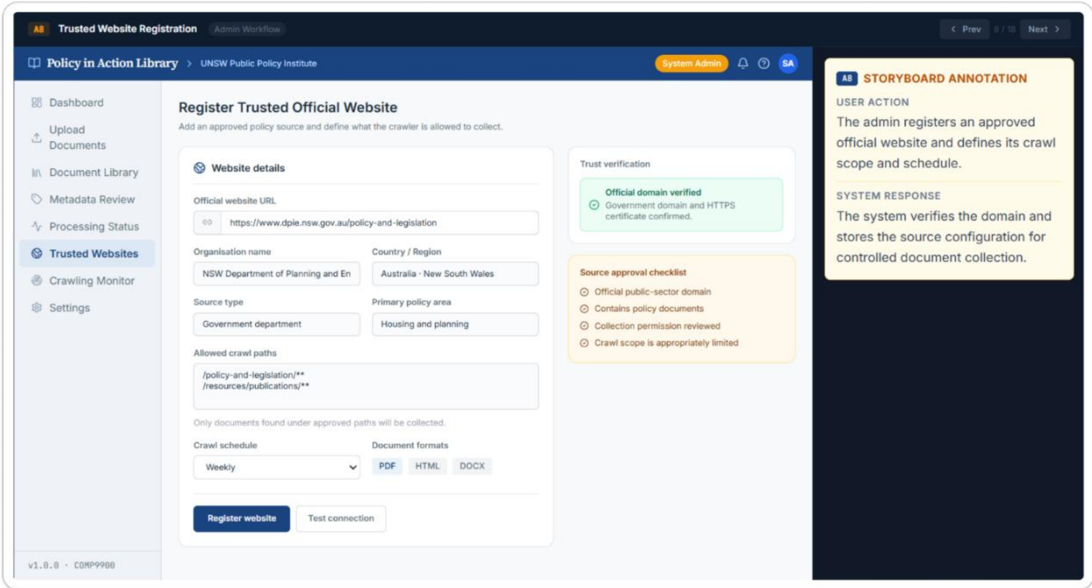
Click the “View” (eye) icon to open the document details.



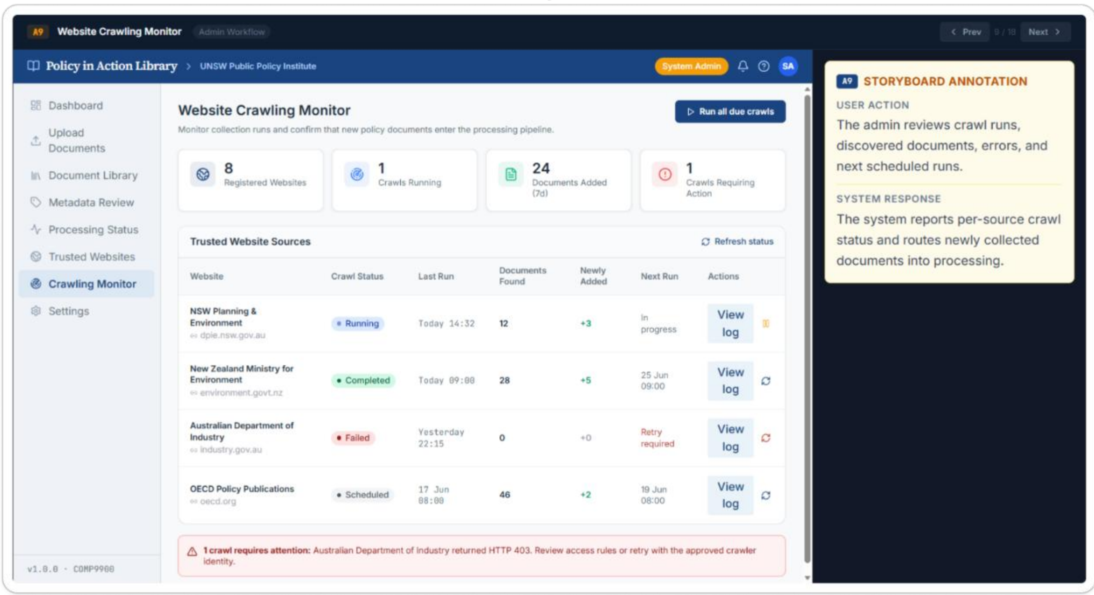
Demonstration: The document management page gives administrators an overview of the files stored in the repository and their processing status (**W11CBREAD-3**). A specific document can be located using its file ID, filename or keywords (**W11CBREAD-4**). Administrators can then update its metadata, reprocess it, exclude it from AI retrieval or permanently delete it when necessary (**W11CBREAD-5**).

1.5 Trusted Website Registration and Monitoring

Covered User Stories: W11CBREAD-22, W11CBREAD-23



Test connection and register the trusted website

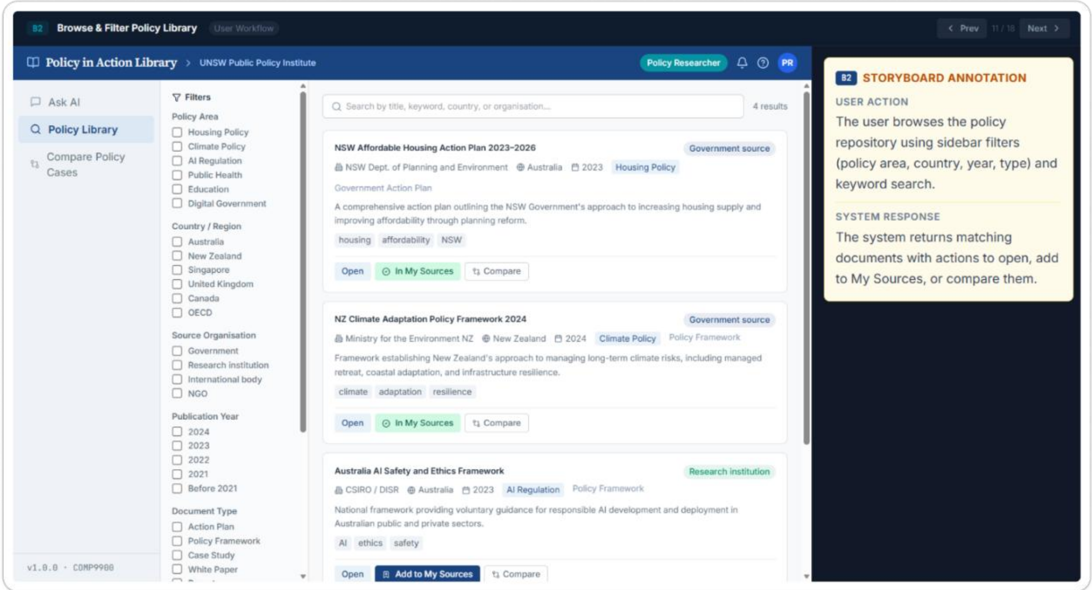


Demonstration: Administrators can register approved official websites as trusted sources and define their permitted paths, collection schedules and supported file formats (**W11CBREAD-22**). The monitoring page shows crawl activity, discovered documents, processing results and failed collection attempts that may need to be retried (**W11CBREAD-23**).

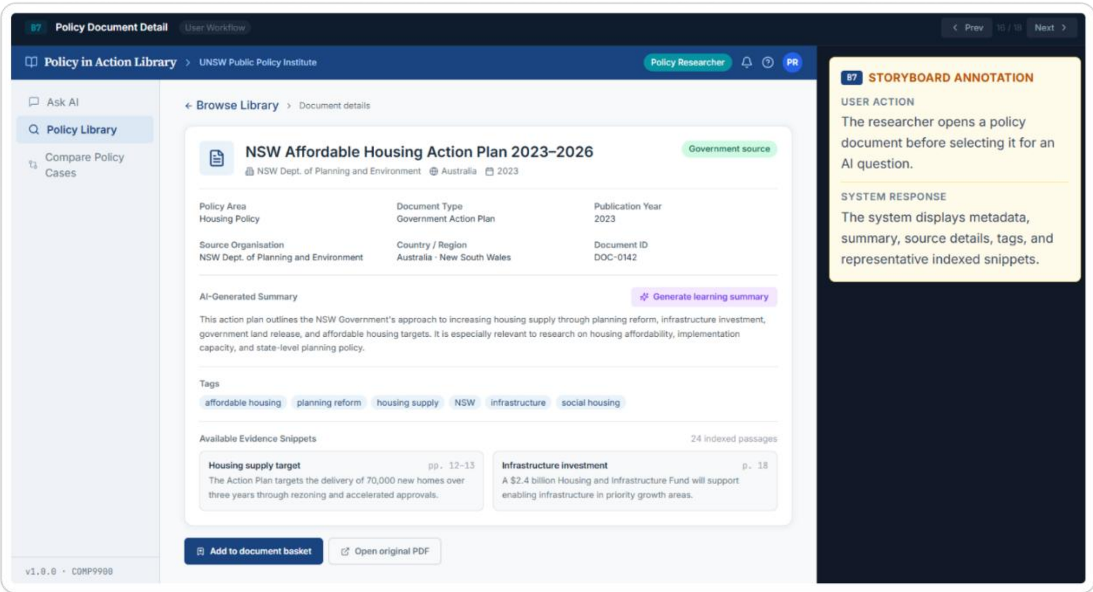
2 Researcher Workflow Storyboards

2.1 Policy Search and Document Review

Covered User Stories: W11CBREAD-14, W11CBREAD-16, W11CBREAD-36



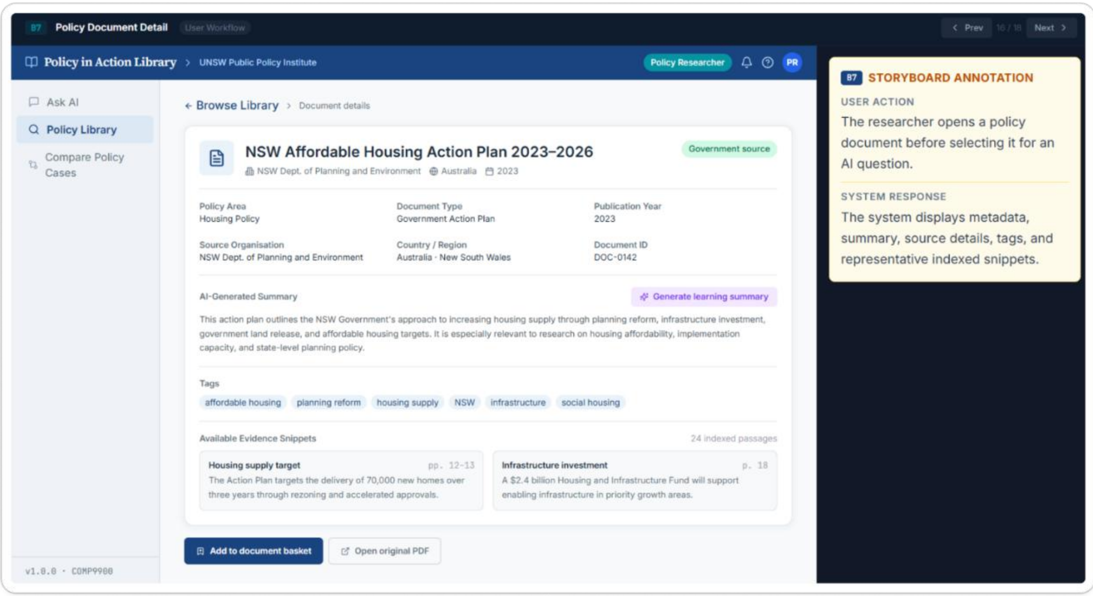
Click "Open" to view the document details



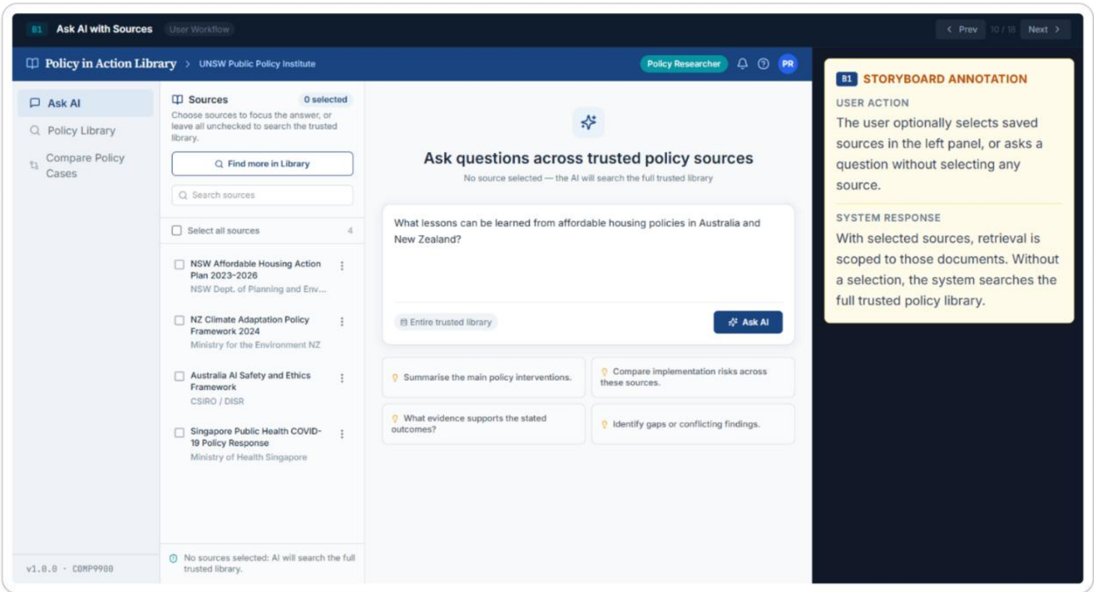
Demonstration: Researchers can locate policy documents by entering a filename or keyword (**W11CBREAD-36**). They can also narrow the results by policy area, country or region, source organisation, publication year and document type (**W11CBREAD-14**). Opening a result displays its metadata, summary, tags, source details and available evidence snippets, helping the researcher decide whether it is relevant (**W11CBREAD-16**).

2.2 Source Selection and Question Submission

Covered User Stories: W11CBREAD-9, W11CBREAD-10



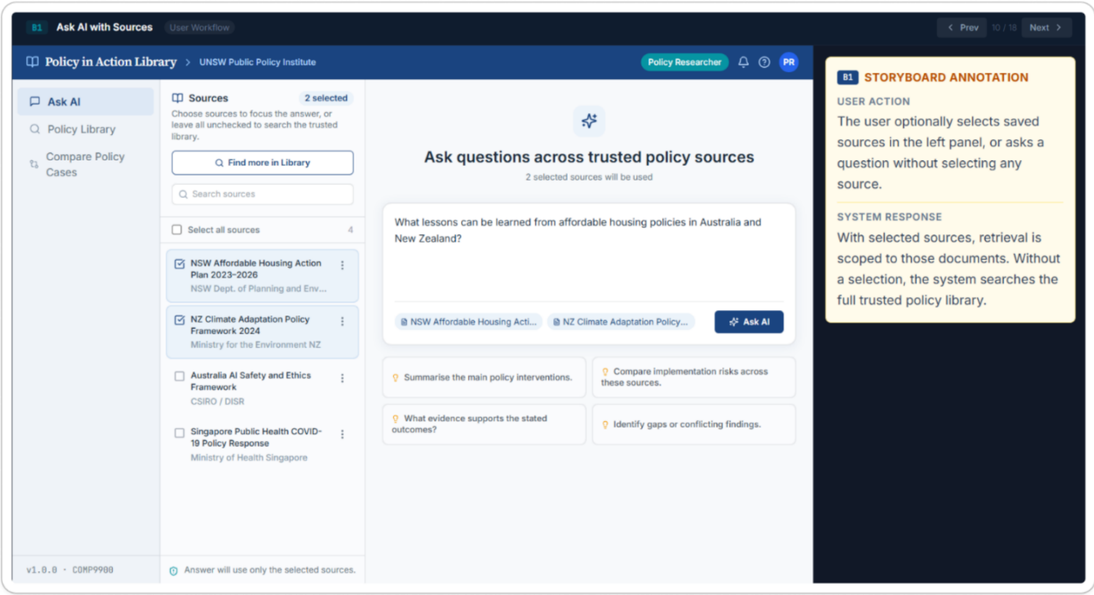
Click "Add to document basket", then continue to "Ask AI"



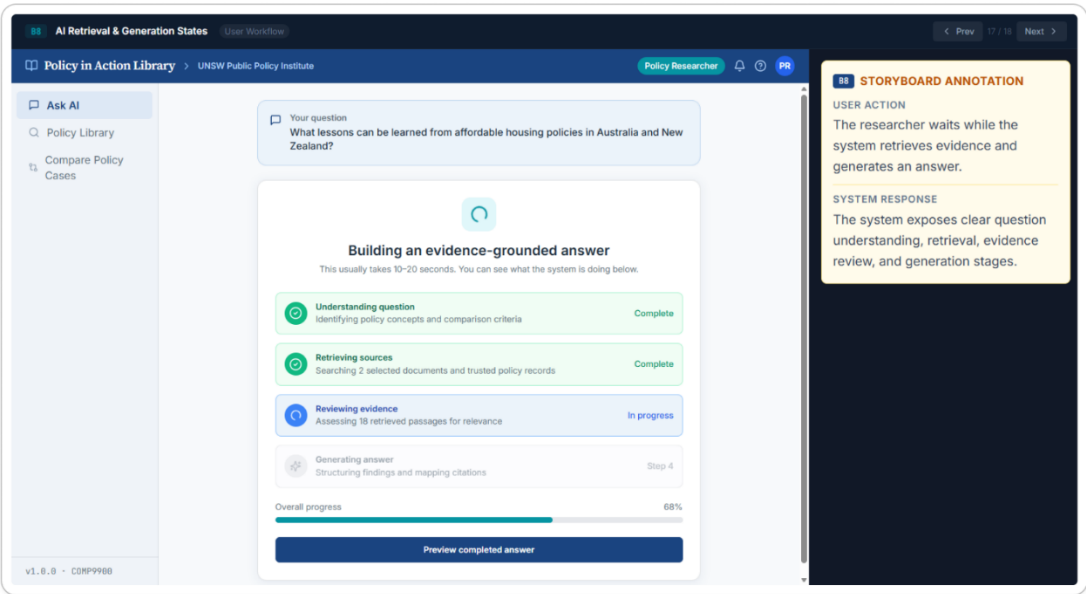
Demonstration: After reviewing a document, the researcher can add it to the document basket. One or more documents may be selected to define the evidence scope used by the AI (**W11CBREAD-10**). The researcher then enters a plain-language question and selects "Ask AI" to begin the analysis (**W11CBREAD-9**).

2.3 AI Answer Generation

Covered User Stories: W11CBREAD-9, W11CBREAD-18



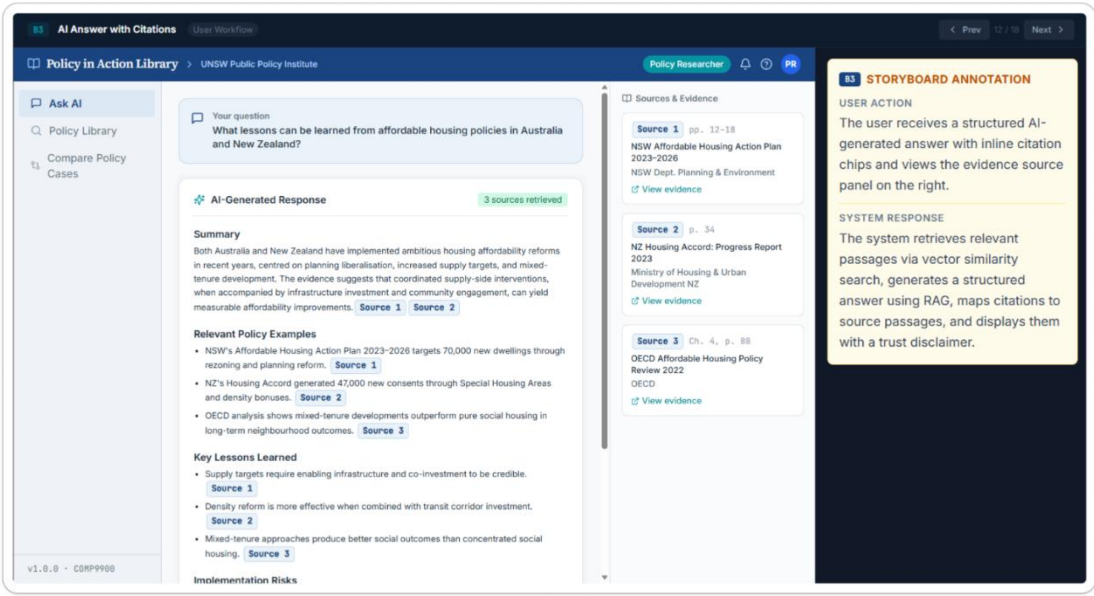
Click “Ask AI” to retrieve evidence and generate an answer



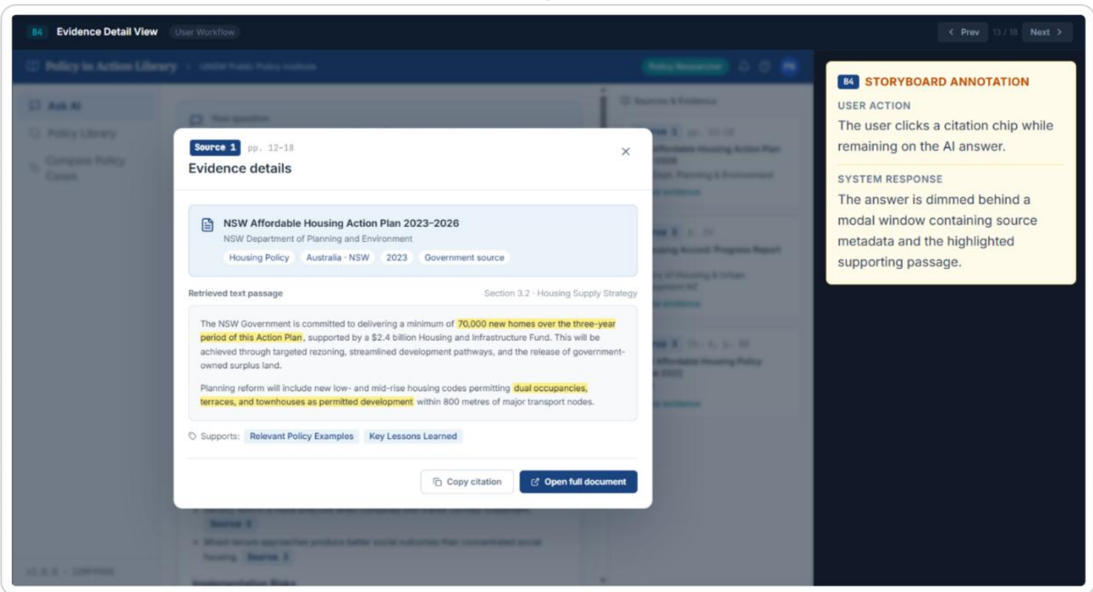
Demonstration: Once the question is submitted, the system begins retrieving and analysing evidence from the selected sources (**W11CBREAD-9**). The page displays the main processing stages—question interpretation, source retrieval, evidence review and answer generation—so the researcher can follow the system's progress (**W11CBREAD-18**).

2.4 AI Answer and Citation Verification

Covered User Stories: W11CBREAD-11, W11CBREAD-19, W11CBREAD-24, W11CBREAD-26



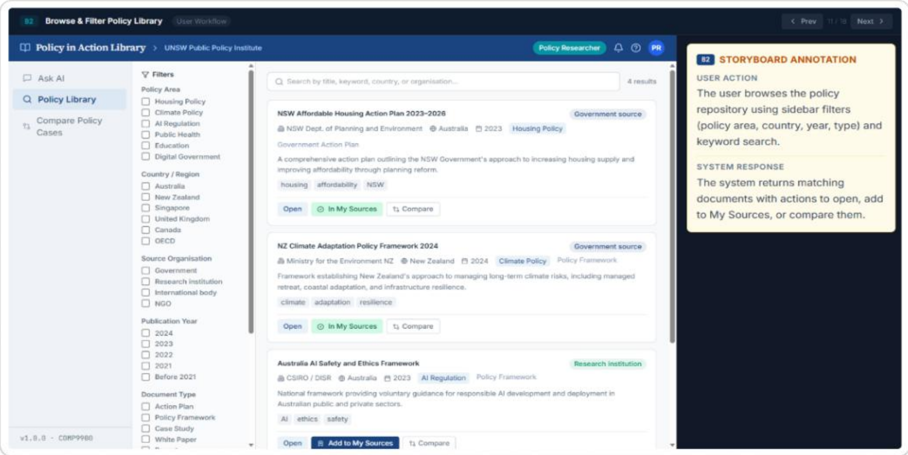
Click a citation to inspect its supporting evidence



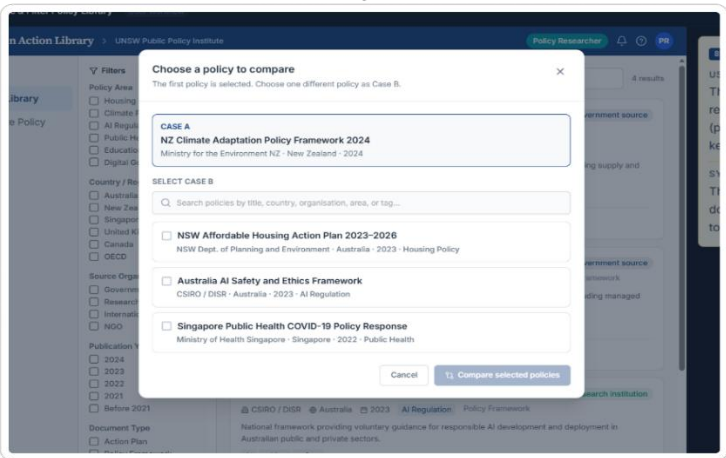
Demonstration: The final response uses clear and accessible language to explain complex policy cases (**W11CBREAD-26**). It is organised into relevant cases, key lessons, risks, implementation considerations and practical recommendations (**W11CBREAD-24**). Citations are included throughout the answer (**W11CBREAD-11**). Selecting an inline citation or “View evidence” displays the relevant document, page and supporting passage without leaving the answer workflow (**W11CBREAD-19**).

2.5 Policy Case Comparison

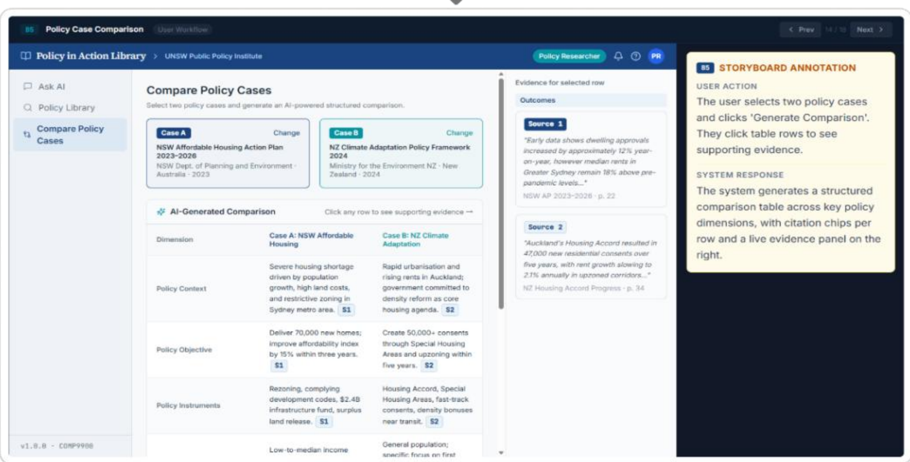
Covered User Stories: W11CBREAD-15, W11CBREAD-35



Click “Compare” to select the first policy case



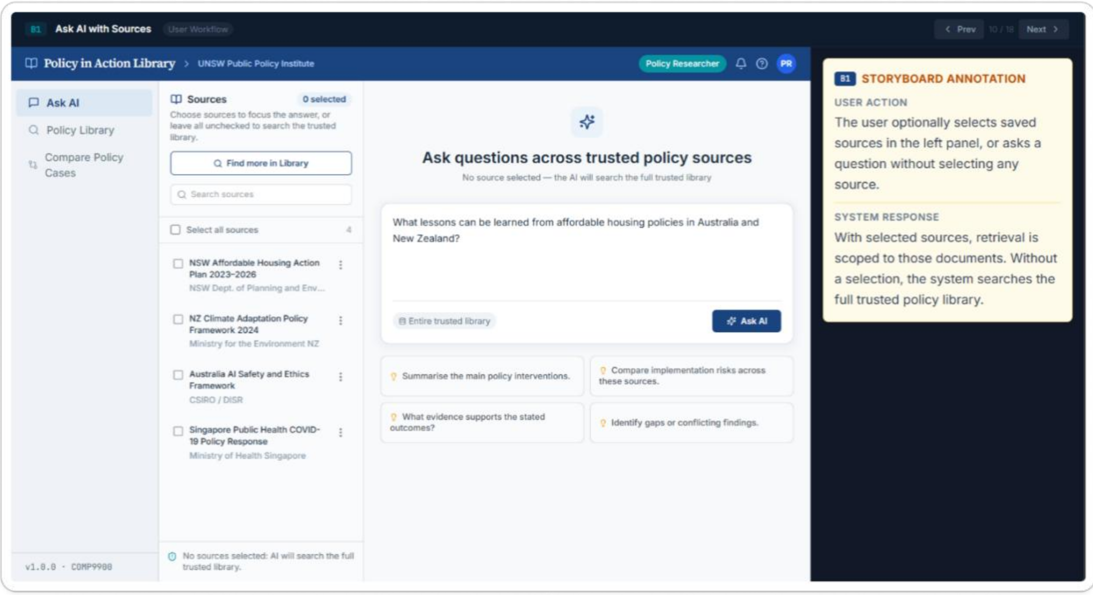
Select Case B and click “Compare selected policies”



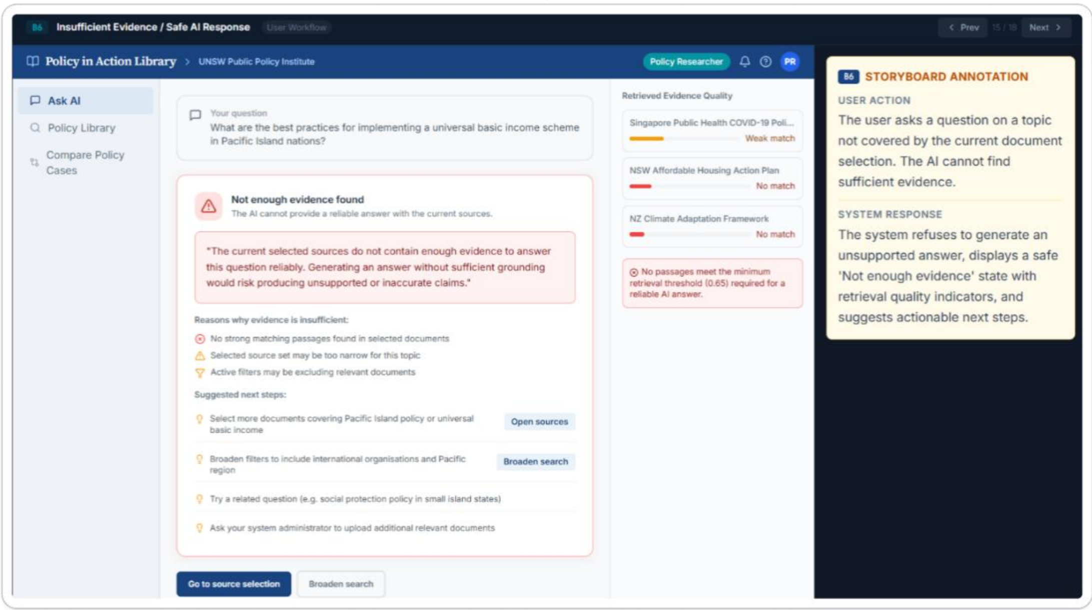
Demonstration: The researcher selects one policy case and then chooses another for comparison (**W11CBREAD-15**). The generated table compares their policy contexts, approaches, outcomes and lessons learned. This helps policy students understand how different policy decisions may lead to different results (**W11CBREAD-35**).

2.6 Insufficient Evidence Handling

Covered User Story: W11CBREAD-13



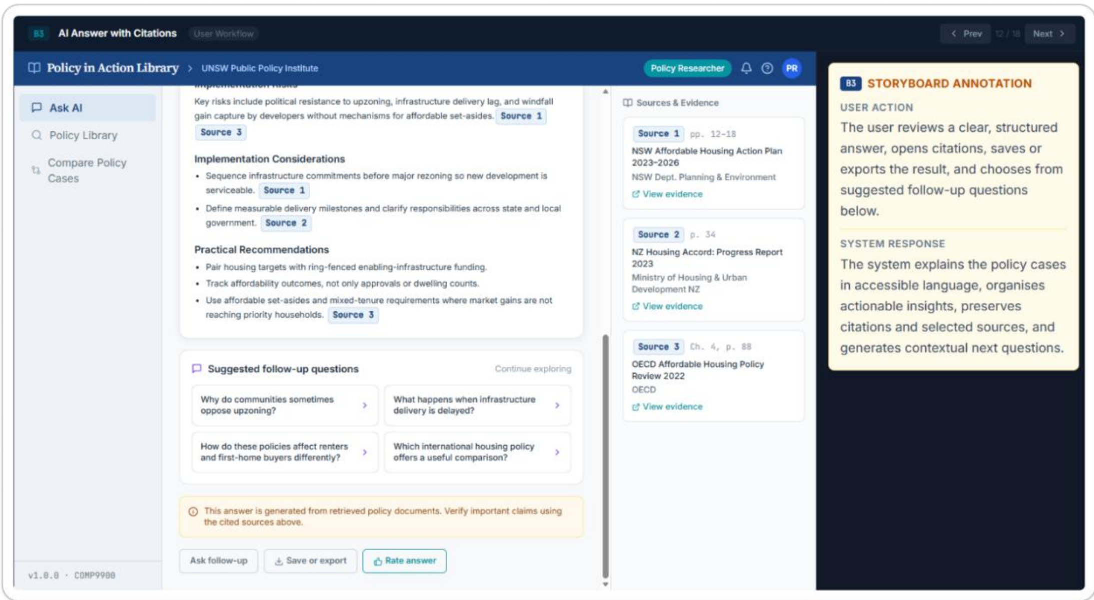
Insufficient evidence detected → Display a safe response



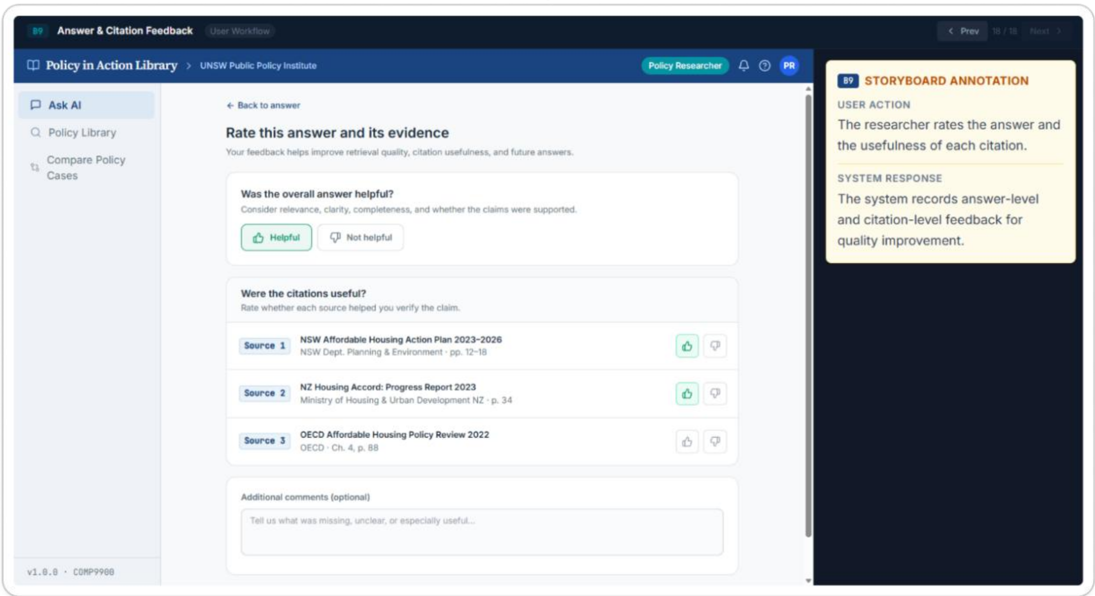
Demonstration: If the retrieved material is not strong enough to support a reliable answer, the system avoids presenting an unsupported conclusion. It explains the limitation and suggests adding more sources, broadening the filters or refining the question (**W11CBREAD-13**).

2.7 Answer and Citation Feedback

Covered User Story: W11CBREAD-20



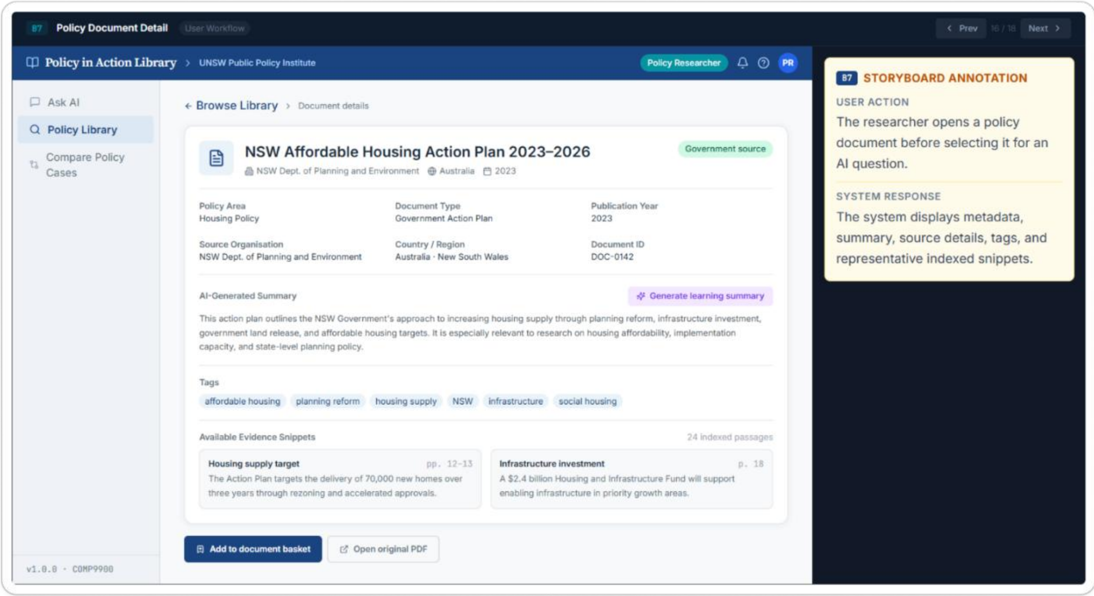
Click “Rate answer” to provide feedback



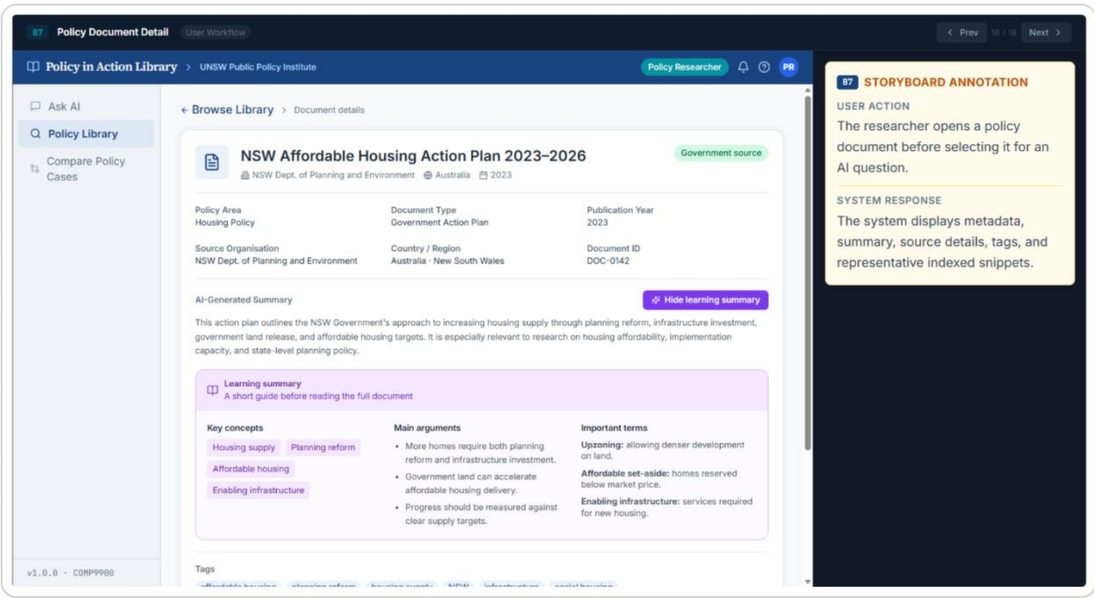
Demonstration: After reviewing an answer, the researcher can rate its relevance, clarity, completeness and supporting evidence. Individual citations can also be rated, with an optional comment box for more detailed feedback. These responses can later support the evaluation of retrieval and answer quality (**W11CBREAD-20**).

2.8 Document Learning Summary

Covered User Story: W11CBREAD-34



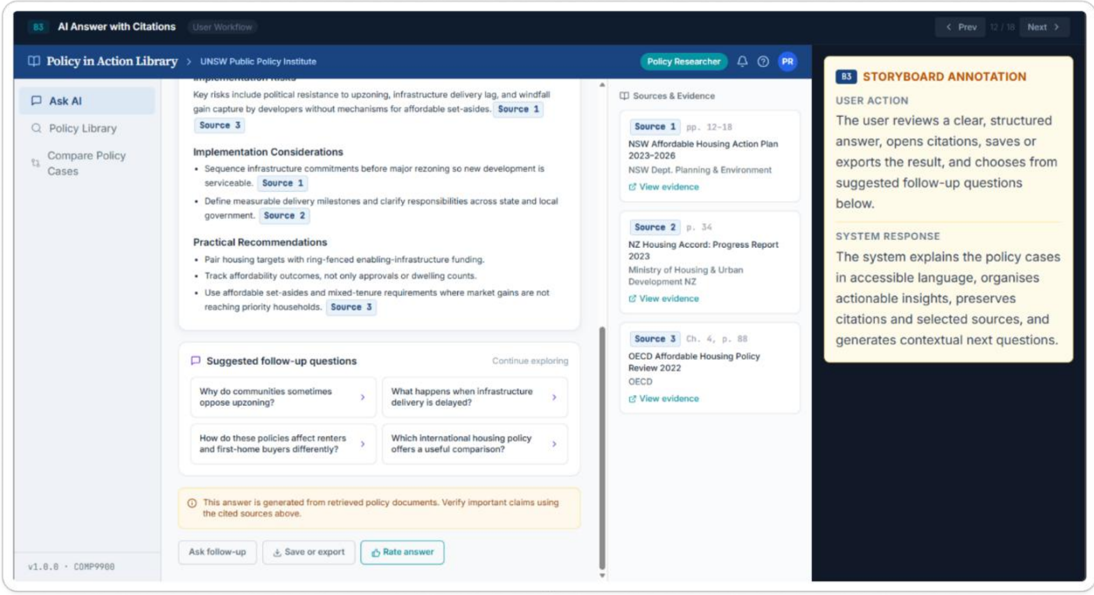
click “Generate learning summary”



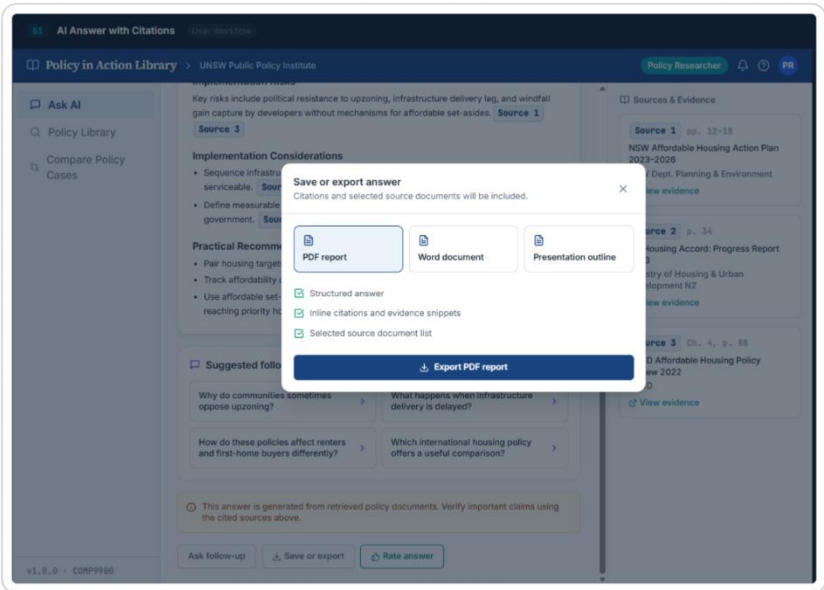
Demonstration: From the document detail page, a policy student can select “Generate learning summary”. The resulting summary highlights the document’s main arguments, key concepts and important policy terms, providing an overview before the student reads the full document (**W11CBREAD-34**).

2.9 Save, Export and Guided Follow-up

Covered User Stories: W11CBREAD-25, W11CBREAD-27



Click “Save or export”



Demonstration: Researchers can save or export an answer together with its citations and selected sources for later use in reports, presentations or further research (**W11CBREAD-25**). Suggested follow-up questions are displayed below the answer, helping policy students explore related examples, policy issues and implementation challenges (**W11CBREAD-27**).

c. Design Justifications

1 Design Requirements Derived from User Stories
The user stories define three main groups: administrators who curate the knowledge base, policy researchers who require evidence-grounded analysis, and policy students who require accessible summaries and comparisons.
These requirements motivate the layered design below. Each layer has a focused responsibility and can be replaced or extended without redesigning the entire platform.

2 User Interface Layer
The interface contains a researcher workspace and an administration dashboard.

Option	Advantages	Limitations	Decision
HTML/Bootstrap	Lowest initial setup and suitable for static pages.	State management becomes difficult for baskets, filters, chat, citations and upload progress.	Not selected.
Django templates	Simple server-rendered integration.	Less suitable for highly interactive, asynchronous research workflows.	Not selected.
Vue/Nuxt	Comparable component model and good developer experience.	Would add a second ecosystem without a clear project-specific benefit.	Viable alternative.
React + Next.js	Reusable components, routing, layouts and integrated server/client data handling.	More framework concepts than a static site.	Selected for the MVP.

3 API and Authentication Layer
A REST API separates the interface from document processing and RAG services.

Backend option	Strength	Trade-off	Decision
Next.js-only backend	Single TypeScript application and simple deployment.	Python AI and PDF libraries require additional integration.	Not preferred.
Django / Django REST Framework	Mature ORM, admin tools and authentication.	Heavier framework than required for the proposed service-oriented MVP.	Viable alternative.
FastAPI	Python-native AI integration, type-based validation, asynchronous endpoints and OpenAPI documentation.	Admin functions and authentication require explicit implementation.	Selected.

4 Document Management Layer

This layer controls how evidence enters and leaves the active knowledge base. An uploaded PDF is stored as a raw file and registered with a checksum, metadata and processing state. Core metadata includes title, source organisation, jurisdiction, policy area, publication date, document type, tags and approval status. The lifecycle is simplified to uploaded, processing, review required, approved, failed and archived. Only successfully processed and approved documents are searchable or available to the RAG pipeline.

5 Knowledge Processing and Indexing Layer

The processing pipeline is: PDF extraction, cleaning, chunking, metadata binding, embedding and indexing.

Decision area	Alternatives	Selected route and rationale
Chunking	Whole document; fixed-size; section-aware; LLM semantic splitting.	Section-aware chunks with a fixed-size overlap fallback. This preserves policy structure without the cost and variability of LLM splitting.
Embeddings	External embedding API; locally hosted model.	A provider-neutral service interface, initially using an API model for development speed. Local models remain possible when privacy or long-term cost dominates.
Vector index	Dedicated vector database; PostgreSQL with pgvector.	pgvector for the MVP because vectors and metadata filters remain in one database. A dedicated service is justified only at substantially larger scale.

6 RAG and AI Reasoning Layer

The expected questions include factual evidence requests, document summaries, cross-country case comparisons, implementation lessons and questions that cannot be answered from the current library. The selected architecture is therefore a metadata-aware RAG pipeline rather than direct LLM question answering. Its main flow is query interpretation, scope filtering, retrieval, reranking, evidence grouping, prompt assembly, generation and citation validation.

6.1 Query, Scope and Retrieval

The system first applies the user's selected documents and approval constraints, then extracts explicit metadata such as countries, years and policy areas.

Retrieval route	Advantages	Limitations	Use in this project
Keyword/BM25	Strong for exact policy names, legislation, acronyms and	Misses conceptually related wording.	Included as a complementary signal.

	agencies.		
Vector search	Finds semantic matches across different terminology.	May underweight exact names and can return semantically broad passages.	Primary retrieval method.
Hybrid retrieval	Combines semantic recall with exact-term precision.	Requires score fusion and evaluation.	Selected overall route.

6.2 Reranking and Evidence Construction

Reranking removes weak or repetitive candidates before prompt construction. Pure similarity-score ordering is cheapest but can overrepresent one document.

6.3 Prompting, Quality Controls and Cost

A small set of prompt templates is used for question answering, summarisation and comparison. Each template instructs the model to use only supplied evidence, cite substantive claims, distinguish reported findings from synthesis, and identify missing evidence. Comparison prompts additionally require balanced coverage, contextual limitations and transferable lessons. Template selection is preferred over an unconstrained agent because it is easier to test and produces more predictable outputs for the MVP.

7 Data, Storage, and Evaluation Layer

7.1 Storage and Schema

The selected storage design combines object or local file storage for original PDFs with PostgreSQL for structured records.

Entity	Key fields	Purpose
documents	id, title, source, country, policy area, dates, file URI, checksum, approval and processing status	Document governance, browsing and filtering.
chunks	id, document ID, text, page range, section, token count	Retrievable evidence and citation traceability.
embeddings	chunk ID, vector, model and version	Semantic search and controlled re-indexing.
queries / citations	question, filters, retrieved chunks, answer, source mappings, latency	Debugging, auditability and evaluation.
feedback	query ID, usefulness, citation rating and comment	User-centred quality monitoring.

d. Functionality Mapping

Requirement	User Story	Justification
Secure administration access	W11CBREAD-8 — Admin login and unauthorised access prevention	Protects restricted administration functions and reduces unauthorised changes to the knowledge base.
Controlled policy document ingestion	W11CBREAD-1 — Upload multiple policy PDFs W11CBREAD-7 — Segment complex PDFs for retrieval	Provides a controlled way to add policy files and prepares their content for retrieval without losing document context.
Policy metadata management	W11CBREAD-2 — Automatic and manual metadata tagging	Keeps documents consistently classified and supports later filtering by country, year, policy area, source and tags.
Administrative document management	W11CBREAD-3 — Browse and filter uploaded files W11CBREAD-4 — Search by ID, filename or keywords W11CBREAD-5 — Delete a policy file	Allows admins to locate, review and remove documents that are duplicated, incorrect or outdated.
Processing status visibility	W11CBREAD-12 — View upload, parsing, segmentation, embedding and indexing status	Shows whether a document is ready for retrieval and gives a clear error state when processing fails.
Trusted external source collection	W11CBREAD-22 — Register trusted official websites W11CBREAD-23 — Monitor website crawling status	Supports future collection from approved public sources while keeping source selection under admin control.
Policy document discovery	W11CBREAD-14 — Browse and filter policy documents W11CBREAD-16 — Open a document detail page W11CBREAD-36 — Search by filename or keywords	Helps users find and review relevant policy materials before deciding which sources to use.
Source-scoped natural-language questioning	W11CBREAD-9 — Ask natural-language questions W11CBREAD-10 — Select one or more documents as the question scope	Lets users ask questions in plain language while keeping the AI response focused on the documents they selected.
AI response progress visibility	W11CBREAD-18 — View loading, retrieval and answer-generation states	Shows what the system is doing while the user waits for document retrieval and answer generation.

Requirement	User Story	Justification
Structured policy analysis	W11CBREAD-24 — Organise responses into cases, lessons, risks, implementation considerations and recommendations	Turns retrieved evidence into a consistent format that is easier to read and use for policy research.
Cross-case comparison	W11CBREAD-15 — Compare policy cases across countries or regions W11CBREAD-35 — Compare two selected policy cases for learning	Supports direct comparison of policy contexts, approaches, outcomes and lessons while retaining source references.
Citation and evidence verification	W11CBREAD-11 — Include citations and source snippets W11CBREAD-19 — Open the supporting source snippet	Allows users to check AI claims against the relevant document and passage.
Insufficient evidence handling	W11CBREAD-13 — Report when the document library does not contain enough evidence	Avoids presenting an unsupported conclusion when the selected sources cannot answer the question reliably.
Answer and citation feedback	W11CBREAD-20 — Rate answer helpfulness and citation usefulness	Provides data that can be used to evaluate and improve retrieval, citations and answer quality.
Student learning support	W11CBREAD-26 — Beginner-friendly explanations W11CBREAD-34 — Short document learning summary W11CBREAD-27 — Guided follow-up questions	Makes complex policy cases easier to understand and supports continued learning from the selected materials.
Saving and exporting results	W11CBREAD-25 — Save or export an answer with its citations and selected sources	Allows users to reuse research results in reports or presentations while keeping the supporting sources attached.

The functionality mapping shows that the main functions of the prototype are directly connected to the user stories. Admin-related functions support the creation and maintenance of the policy knowledge base. User-facing functions support document discovery, source selection, natural-language questioning, and citation-based verification. The RAG-related functions connect these two sides by retrieving relevant evidence from approved documents and generating source-grounded responses.

Overall, the system is not designed as a general chatbot. Its key function is the combination of curated document management, metadata-based discovery, selected-source retrieval, AI synthesis, and citation support. This ensures that the prototype remains aligned with the project goal of helping users obtain traceable and useful information from public policy materials.